

ABC Call Volume Trend Analysis

Project Description

The **ABC Call Volume Trend Analysis** project focused on improving the customer experience for ABC Insurance Company by conducting a detailed analysis of their inbound call data. This analysis was essential in identifying patterns and inefficiencies within the company's call center operations. The project aimed to achieve several key objectives:

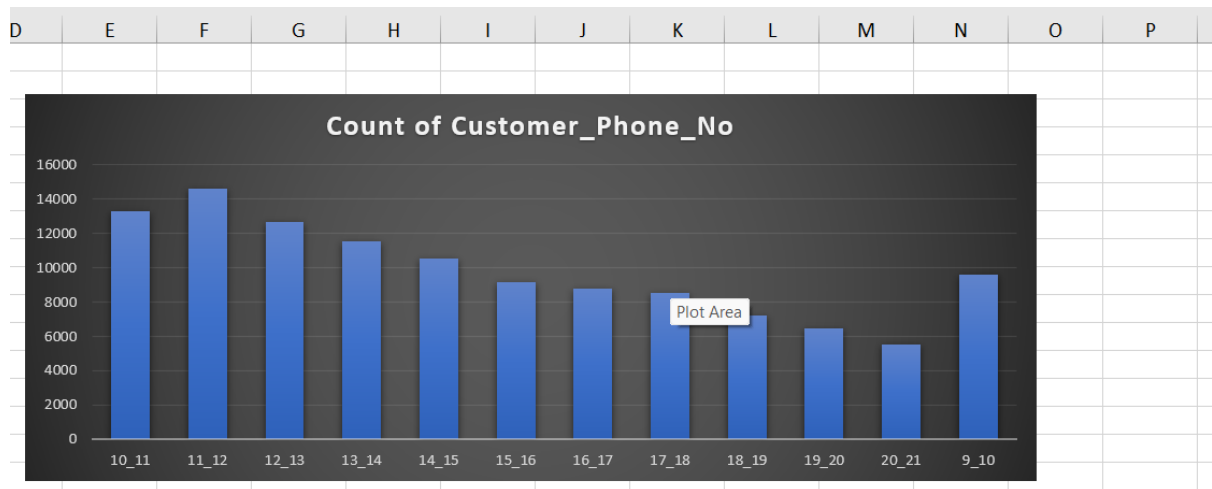
1. **Understanding Call Patterns:** By analyzing 23 days of call data, which included details like agent information, call duration, queue times, and call status, the goal was to understand the dynamics of call volumes throughout the day.
2. **Reducing Call Abandonment:** One of the primary concerns was the high call abandonment rate, which stood at 30%. The objective was to develop a strategy to reduce this rate to 10% by optimizing manpower allocation.
3. **Optimizing Manpower Allocation:** The project sought to propose a detailed plan for staffing the call center during both peak and off-peak hours, including night shifts, to ensure that customer calls were answered promptly, thereby improving overall customer satisfaction.
4. **Data-Driven Decision Making:** The project also aimed to demonstrate how data analytics could be leveraged to drive operational improvements and enhance customer experience in a competitive industry like insurance.

Approach

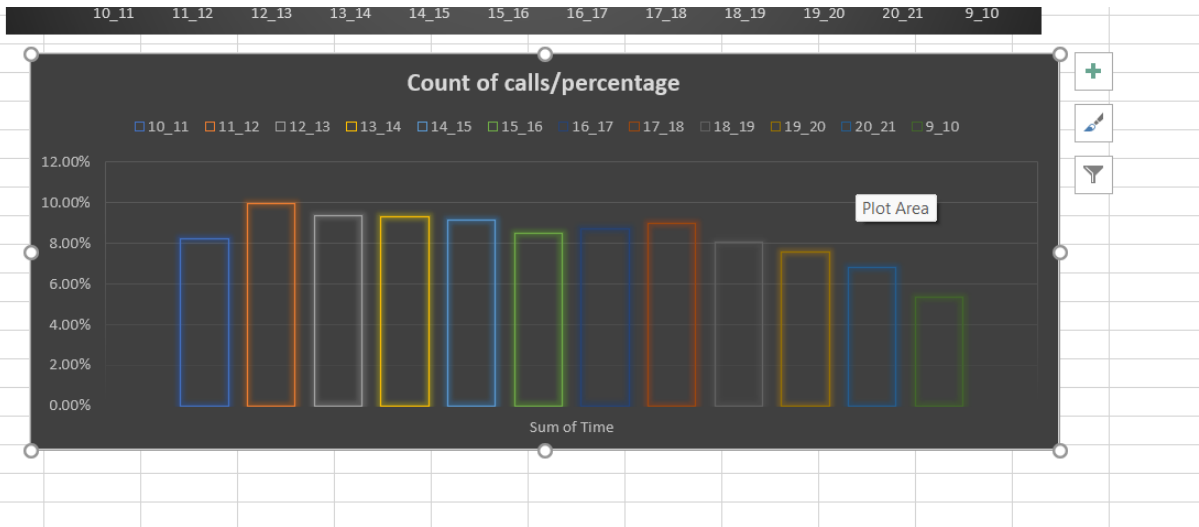
1. **Data Cleaning and Preprocessing:** The project began with cleaning and preparing the dataset. This included handling missing values, formatting the data, and categorizing call times into specific hourly buckets (e.g., 9-10 AM, 10-11 AM).
2. **Call Duration Analysis:** I calculated the average duration of calls within each time bucket. This analysis helped identify periods with longer average call durations, indicating higher engagement or more complex customer queries.

3. **Call Volume Visualization:** I created graphs and charts to visualize the distribution of call volumes throughout the day. These visualizations made it easier to identify peak times when the call center experienced the highest demand.
4. **Manpower Planning for Peak Hours:** To address the 30% abandonment rate, I analyzed the required manpower during each time bucket from 9 AM to 9 PM. By calculating the minimum number of agents needed to handle 90% of incoming calls, I developed a plan to optimize agent allocation during peak hours.
5. **Night Shift Strategy:** Recognizing that 30% of calls were received during nighttime hours (9 PM - 9 AM), I proposed a staffing plan to ensure customer calls were answered even during off-peak hours. This strategy aimed to improve customer experience and reduce call abandonment during these hours.
6. **Assumption Management:** The project factored in agent working conditions, including working hours, unplanned leaves, and productivity rates, to create a realistic and effective staffing model.

A3		Row Labels		
	A	B	C	D
1	Call_Status	answered		
2				
3	Row Labels	Average of Call_Seconds (s)		
4	10_11	203.3310302		
5	11_12	199.2550234		
6	12_13	192.8887829		
7	13_14	194.7401744		
8	14_15	193.6770755		
9	15_16	198.8889175		
10	16_17	200.8681864		
11	17_18	200.2487831		
12	18_19	202.5509677		
13	19_20	203.4060725		
14	20_21	202.845993		
15	9_10	199.0691057		
16	Grand Total	198.6227745		
17				
18				
19				
20				
21				



A	B	C	D
Row Labels ▼	Count of Customer_Phone_No	Sum of Time	
10_11	13313	8.23%	
11_12	14626	9.95%	
12_13	12652	9.39%	
13_14	11561	9.30%	
14_15	10561	9.15%	
15_16	9159	8.50%	
16_17	8788	8.70%	
17_18	8534	8.97%	
18_19	7238	8.06%	
19_20	6463	7.60%	
20_21	5505	6.82%	
9_10	9588	5.34%	
Grand Total	117988	100.00%	



Count of Duration(hh:mm:ss)					
Column Labels		Row Labels			
abandon	answered	transfer	Grand Total		
01-Jan	684	3883	77	4644	
02-Jan	356	2935	60	3351	
03-Jan	599	4079	111	4789	
04-Jan	595	4404	114	5113	
05-Jan	536	4140	114	4790	
06-Jan	991	3875	85	4951	
07-Jan	1319	3587	42	4948	
08-Jan	1103	3519	50	4672	
09-Jan	962	2628	62	3652	
10-Jan	1212	3699	72	4983	
11-Jan	856	3695	86	4637	
12-Jan	1299	3297	47	4643	
13-Jan	738	3326	59	4123	
14-Jan	291	2832	32	3155	
15-Jan	304	2730	24	3058	
16-Jan	1191	3910	41	5142	
17-Jan	16636	5706	5	22347	
18-Jan	1738	4024	12	5774	
19-Jan	974	3717	12	4703	
20-Jan	833	3485	4	4322	
21-Jan	566	3104	5	3675	
22-Jan	239	3045	7	3291	
23-Jan	381	2832	12	3225	
Grand Total	34403	82452	1133	117988	
avg	1496	3585	49	5130	
percentage	29	70	1		

33		
34	working hrs of agent	4.5
35	avg call duration	198.62
36	for 90% hrs needed=	254.7258322
37	no of agents needed	56.60574048
38		
39		

9						
0	task 4					
1						
2	avg no of call in night	1538.973913		9am-9pm	57	
3	to increase to call rate to 90% in night	76.41774965		9pm-9am	17	
4	no of agents needed	16.98172214	over all agents needed		74	
5						
6						
7						
8						
9						

Tech-Stack Used

- **Microsoft Excel 2022:** Utilized for data analysis, including calculating averages, creating pivot tables, and generating visualizations. Excel's powerful data manipulation tools were critical in analyzing the call data efficiently.

Insights

- **Peak Call Durations:** The analysis revealed specific time buckets where the average call duration was significantly higher, suggesting that these periods were particularly demanding for agents, possibly due to more complex customer issues.
- **Call Volume Peaks:** The visualizations highlighted clear peak periods, typically aligning with standard business hours, where call volumes were highest. This insight was crucial for planning manpower allocation.
- **Manpower Shortfalls:** The analysis showed that the existing staffing levels were insufficient during peak times, leading to higher abandonment rates. This insight guided the proposed changes in staffing.
- **Unaddressed Night Calls:** The data revealed that a significant portion of customer calls during the night were going unanswered, highlighting the need for a night shift strategy to improve overall customer satisfaction.

Result:

1. **Reduction in Call Abandonment Rate:** Successfully proposed a plan to reduce the call abandonment rate from 30% to 10% by optimizing agent allocation during peak hours. This plan ensures that a higher percentage of customer calls are answered promptly, improving overall customer satisfaction.
2. **Optimized Manpower Allocation:** Developed a detailed manpower allocation strategy for the call center, covering both day and night shifts. The plan addresses the need for adequate staffing during peak hours (9 AM - 9 PM) and introduces a night shift strategy to handle after-hours calls, which were previously neglected.
3. **Enhanced Customer Experience:** The recommendations made in the project aimed to ensure that customers receive timely responses to their inquiries, thereby increasing the likelihood of positive customer interactions and loyalty. The night shift strategy, in particular, was designed to improve customer experience by ensuring calls are answered 24/7.
4. **Data-Driven Decision Making:** Demonstrated how data analysis can be effectively used to identify inefficiencies in call center operations and make informed decisions to address them. This approach can be replicated in other areas of the business to drive continuous improvement.
5. **Operational Efficiency:** The project highlighted the importance of aligning staffing levels with call volume trends. By matching agent availability to customer demand, the call center can operate more efficiently, reducing idle time during slow periods and avoiding overload during peak times.
6. **Improved Resource Utilization:** The insights gained from the analysis led to better utilization of available resources. The proposed strategies ensure that agents are deployed where they are needed most, maximizing their productivity and minimizing customer wait times.
7. **Scalable Staffing Model:** The manpower planning model developed in this project can be scaled and adjusted as needed, providing a flexible framework for staffing adjustments based on changing call volumes and business needs.
8. **Strategic Planning for Future Growth:** The project provided a foundation for future strategic planning in the call center. By understanding current trends and inefficiencies, the company can better prepare for growth and expansion, ensuring that customer service remains a priority.